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QUICK CONNECT CONFIGURATIONS FOR WELDING NECKS AND GAS DIFFUSERS

Abstract

Some examples of the present disclosure relate to apparatus, systems, and/or methods for providing a quick connect and/or disconnect for a gas diffuser and/or neck assembly, for example, in a welding system. The gas diffuser may include threaded grooves and/or protrusions configured to engage with screw threads and/or channels of the neck assembly. The screw threads, protrusions, threaded grooves, and/or channels may be configured such that the gas diffuser may be quickly connected to, and/or disconnected from, the neck assembly.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of, and claims priority to, co-pending U.S. Non-Provisional patent application Ser. No. 17/388,219, entitled “QUICK CONNECT CONFIGURATIONS FOR WELDING NECKS AND GAS DIFFUSERS,” filed Jul. 29, 2021, which is a continuation of, and claims priority to, U.S. Non-Provisional patent application Ser. No. 15/944,588 (now U.S. Pat. No. 11,103,949), entitled “QUICK CONNECT CONFIGURATIONS FOR WELDING NECKS AND GAS DIFFUSERS,” filed Apr. 3, 2018 (issued Aug. 31, 2021), which claims priority to, and the benefit of, U.S. Provisional Patent Application No. 62/480,912, entitled “WELDING NECK AND GAS DIFFUSER QUICK CONNECT,” having a filing date of Apr. 3, 2017, all of which are incorporated herein by reference in their entirety.

TECHNICAL FIELD
[0002] The present disclosure generally relates to welding torches, and, more particularly, to quick connect and/or disconnect configurations for welding necks and/or gas diffusers.
BACKGROUND
[0003] Welding torches sometimes include gas diffusers to help route shielding gas to a weld. In some cases, the gas diffusers may be connected to a neck of the welding torch via a screw connection. Gas diffusers may be screwed onto and/or off of the necks when assembling and/or disassembling welding torches.
[0004] Limitations and disadvantages of conventional and traditional approaches will become apparent to one of skill in the art, through comparison of such systems with the present disclosure as set forth in the remainder of the present application with reference to the drawings.
SUMMARY
[0005] The present disclosure is directed to apparatuses, systems, and methods for providing a quick connect and/or disconnect for gas diffusers and/or welding necks, such as in a welding system, for example, substantially as illustrated by and/or described in connection with at least one of the figures, and as set forth more completely in the claims.
[0006] These and other advantages, aspects and novel features of the present disclosure, as well as details of an illustrated example thereof, will be more fully understood from the following description and drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0007] FIG. **1** is an example of a welding system, in accordance with aspects of this disclosure.
[0008] FIG. **2** is a side view of an example welding torch used the welding system of FIG. **1**, in accordance with aspects of this disclosure.
[0009] FIG. **3**A is a perspective view of an example neck and nozzle assembly that may be used in the example welding torch of FIG. **2**, in accordance with aspects of this disclosure.
[0010] FIG. **3**B is a side view of the example neck and nozzle assembly of FIG. **3**A, in accordance with aspects of this disclosure.
[0011] FIG. **4**A is an exploded view of the example neck and nozzle assembly of FIG. **3**A, in

accordance with aspects of this disclosure.
[0012] FIG. **4**B is an exploded view of an example neck assembly of the neck and nozzle assembly of FIG. **4**A, in accordance with aspects of this disclosure.
[0013] FIG. **4**C is an exploded view of an example nozzle assembly of the neck and nozzle assembly of FIG. **4**A, in accordance with aspects of this disclosure.
[0014] FIG. **5**A is a side view of an example gas diffuser, in accordance with aspects of this disclosure.
[0015] FIG. **5**B is an end view of the example gas diffuser of FIG. **5**A, in accordance with aspects of this disclosure.
[0016] FIG. **5**C is a cross section of the example gas diffuser of FIG. **5**A, along line **5**C-**5**C of FIG. **5**A, in accordance with aspects of this disclosure.
[0017] FIG. **5**D is an enlarged view of threaded grooves of the example gas diffuser of FIG. **5**C, in accordance with aspects of this disclosure.
[0018] FIG. **6**A is an end view of an example neck inner portion, in accordance with aspects of this disclosure.
[0019] FIG. **6**B is a side view of the example neck inner portion of FIG. **6**A, in accordance with aspects of this disclosure.
[0020] FIGS. **6**C and **6**D are enlarged views of screw threads of the neck inner portion of FIG. **6**B, in accordance with aspects of this disclosure.
[0021] FIG. **7** is a table showing various dimensions of the example threaded grooves of FIG. **5**D and example screw threads of FIG. **6**C, in accordance with aspects of the present disclosure.
[0022] FIG. **8**A is a perspective view of the example neck and nozzle assembly of FIG. **3**A with a portion cutaway, in accordance with aspects of the present disclosure.
[0023] FIG. **8**B is a cross-sectional view of the example neck and nozzle assembly of FIG. **3**A, along the line **8**B-**8**B of FIG. **3**B, in accordance with aspects of the present disclosure.
[0024] FIG. **9**A is a cross-sectional view of an example neck and nozzle assembly in which an example gas diffuser and example neck assembly have a face-to-face bearing surface, in accordance with aspects of the present disclosure.
[0025] FIG. **9**B shows a cross-sectional view of an example neck and nozzle assembly having custom double start ANSI-style screw threads and threaded grooves, in accordance with aspects of the present disclosure.
[0026] FIG. **10**A is a side view of another example neck inner portion, in accordance with aspects of this disclosure.
[0027] FIG. **10**B is an end view of another example gas diffuser, in accordance with aspects of this disclosure.

DETAILED DESCRIPTION

[0028] Preferred examples of the present disclosure may be described hereinbelow with reference to the accompanying drawings. In the following description, well-known functions or constructions are not described in detail because they may obscure the disclosure in unnecessary detail. For this disclosure, the following terms and definitions shall apply.
[0029] As utilized herein, “and/or” means any one or more of the items in the list joined by “and/or”. As an example, “x and/or y” means any element of the three-element set {(x), (y), (x, y)}. In other words, “x and/or y” means “one or both of x and y”. As another example, “x, y, and/or z” means any element of the seven-element set {(x), (y), (z), (x, y), (x, z), (y, z), (x, y, z)}. In other words, “x, y and/or z” means “one or more of x, y and z”.
[0030] As utilized herein, the term “exemplary” means serving as a non-limiting example, instance, or illustration. As utilized herein, the terms “e.g.” and “for example” set off lists of one or more non-limiting examples, instances, or illustrations.
[0031] As used herein, a welding-type power supply and/or power source refers to any device capable of, when power is applied thereto, supplying welding, cladding, plasma cutting, induction

heating, laser (including laser welding, laser hybrid, and laser cladding), carbon arc cutting or gouging and/or resistive preheating, including but not limited to transformer-rectifiers, inverters, converters, resonant power supplies, quasi-resonant power supplies, switch-mode power supplies, etc., as well as control circuitry and other ancillary circuitry associated therewith.

[0032] Welding-type power, as used herein, refers to power suitable for welding, cladding, plasma cutting, induction heating, CAC-A and/or hot wire welding/preheating (including laser welding and laser cladding), carbon arc cutting or gouging, and/or resistive preheating.

[0033] The terms “coupled,” “coupled to,” and “coupled with” as used herein, each mean a structural and/or electrical connection, whether attached, affixed, connected, joined, fastened, linked, and/or otherwise secured. As used herein, the term “attach” means to affix, couple, connect, join, fasten, link, and/or otherwise secure. As used herein, the term “connect” means to attach, affix, couple, join, fasten, link, and/or otherwise secure.

[0034] The terms “about” and/or “approximately,” when used to modify or describe a value (or range of values), position, orientation, and/or action, mean reasonably close to that value, range of values, position, orientation, and/or action. Thus, the examples described herein are not limited to only the recited values, ranges of values, positions, orientations, and/or actions but rather should include reasonably workable deviations.

[0035] As used herein, the terms “front” and/or “forward” refer to locations closer to a welding arc, while “rear,” “behind,” and/or “backward” refers to locations farther from a welding arc.

[0036] The term “power” is used throughout this specification for convenience, but also includes related measures such as energy, current, voltage, and enthalpy. For example, controlling “power” may involve controlling voltage, current, energy, and/or enthalpy, and/or controlling based on “power” may involve controlling based on voltage, current, energy, and/or enthalpy.

[0037] Some examples of the present disclosure relate to a welding torch, comprising a neck assembly having a protrusion, and a gas diffuser having a passage configured to interface with the protrusion to connect the gas diffuser to the neck assembly in approximately 1.5 turns or less.

[0038] In some examples, the protrusion is a screw thread, and the passage is a threaded groove. In some examples, the screw thread and the threaded groove are based on a custom double-start stub ACME thread. In some examples, the neck assembly includes a locking taper that is configured to interface with a complementary taper of the gas diffuser. In some examples, the screw thread comprises a first thread and a second thread, and the threaded groove comprises a first groove configured to interface with the first thread, and a second groove configured to interface with the second thread. In some examples, the screw thread has a total axial length that is approximately 1.5 times or less an axial lead length of the screw thread. In some examples, the threaded groove has a total axial length that is approximately 1.5 times or less an axial lead length of the threaded groove. In some examples, the passage comprises an axially (and/or longitudinally) extending keyway that leads to a radially (and/or circumferentially, circularly) extending channel.

[0039] Some examples of the present disclosure relate to a neck assembly for use with a welding torch, comprising a neck inner having a protrusion that is configured to interface with a passage of a gas diffuser, where the protrusion is configured to remove the gas diffuser from the neck inner in approximately 1.5 turns or less.

[0040] In some examples, the protrusion is a screw thread. In some examples, the screw thread is based on a custom double-start stub ACME thread. In some examples, the screw thread comprises a first thread and a second thread. In some examples, the screw thread has a total axial length that is approximately 1.5 times or less an axial lead length of the screw thread. In some examples, the neck assembly includes a locking taper and an insulator, and the protrusion is disposed between the locking taper and the insulator. In some examples, the neck assembly further includes an outer neck armor, the outer neck armor surrounds the insulator, and at least part of the neck inner is positioned within the insulator.

[0041] Some examples of the present disclosure relate to a gas diffuser for use in a welding torch, comprising a gas diffuser housing having a passage that is configured to interface with a protrusion of a neck assembly, where the passage is configured to connect the gas diffuser to, or disconnect the gas diffuser from, the neck assembly in approximately 1.5 turns or less.

[0042] In some examples, the passage comprises a threaded groove. In some examples, the threaded groove is based on a custom double-start stub ACME thread. In some examples, the threaded groove comprises a first groove and a second groove. In some examples, the threaded groove has a total axial length that is approximately 1.5 times or less an axial lead length of the threaded groove. In some examples, the gas diffuser includes a taper that is configured to interface with a locking taper of the neck assembly.

[0043] Some examples of the present disclosure relate to apparatuses, systems, and/or methods that provide a quick connect to, and/or a quick disconnect from, a gas diffuser and/or welding torch neck. Gas diffusers and/or welding torch necks may be used, for example, in a welding torch assembly of a welding system. In some examples, a custom thread design for an end of the welding torch neck (e.g., a MIG welding gun neck) is provided for quick assembly and/or removal of the gas diffuser from the neck.

[0044] Conventional welding gun necks can include, for example, a standard tapered pipe thread or a standard thread form such as ISO, ANSI, etc. These types of threads may require multiple turns of the gas diffuser by the user when assembling (i.e., connecting) and/or removing (i.e., disconnecting) a gas diffuser from a neck of a welding torch (and/or welding gun).

[0045] Some examples of the present disclosure provide a custom-sized double-start stub ACME thread that provides for the gas diffuser in a welding torch to be assembled (i.e., connected) or removed (i.e., disconnected, disassembled, etc.) from a welding torch neck with approximately 1.5 turns or less of the gas diffuser and/or welding torch neck (e.g., 1.3, 1.4, 1.5, 1.6, or 1.7 turns). In some examples, the custom thread may be configured to allow assembly/disassembly from a welding torch neck with approximately 2 turns (or less) of the gas diffuser and/or welding torch neck (e.g., 1.8, 1.9, 2.0, 2.1, or 2.2 turns). This substantially reduces the amount of time required to assemble (i.e., connect) and/or remove (i.e., disconnect) a gas diffuser. The customization of the double-start stub ACME thread relates to a custom-sized major diameter, minor diameter, pitch diameter, thread lead, and/or thread pitch.

[0046] Some examples of the present disclosure provide that the sizing of the custom stub ACME thread (e.g., diameters, pitch, etc.) can vary depending on the size of the welding torch neck (e.g., welding gun neck) and the mating gas diffuser. The custom thread can also be constructed by using another standard thread such as, for example, ISO, ANSI, etc. This thread can be used in combination with a locking taper as shown in U.S. Pat. No. 7,176,412, which is hereby incorporated by reference in its entirety, or with a face-to-face mating connection.

[0047] FIG. **1** shows an example of a welding-type system **10**. While the specific welding-type system **10** of FIG. **1** is a gas metal arc welding (GMAW) system, other types of welding-type systems may be used. FIG. **1** illustrates the welding-type system **10** as including a welding-type power source **12** coupled to a wire feeder **14**, though, in some examples, the wire feeder **14** may be removed from the system **10**. In the example of FIG. **1**, the power source **12** supplies welding-type power to a torch **16** through the wire feeder **14**. In some examples, the power source **12** may supply welding-type power directly to the torch **16** rather than through the wire feeder **14**. In the example of FIG. **1**, the wire feeder **14** supplies a wire electrode **18** (e.g., solid wire, cored wire, coated wire) to the torch **16**. A gas supply **20**, which may be integral with or separate from the power source **12**, supplies a gas (e.g., CO2, argon) to the torch **16**. In some examples, no gas supply **20** may be used. An operator may engage a trigger **22** of the torch **16** to initiate an arc **24** between the electrode **18** and a work piece **26**. In some examples, engaging the trigger **22** of the torch **16** may initiate a different welding-type function, instead of an arc **24**.

[0048] In some examples, the welding system **10** may receive weld settings from the operator via an operator interface **28** provided on the power source **12** (and/or power source housing). The weld settings may be communicated to control circuitry **30** within the power source **12** that controls generation of welding-type power for carrying out the desired welding-type operation. In the example of FIG. **1**, the control circuitry **30** is coupled to the power conversion circuitry **32**, which may supply the welding-type power (e.g., pulsed waveform) that is applied to the torch **16**. In the example of FIG. **1**, the power conversion circuitry **32** is coupled to a source of electrical power as indicated by arrow **34**. The source may be a power grid, an engine-driven generator, batteries, fuel cells or other alternative sources.

[0049] In some examples, the control circuitry **30** may control the current and/or the voltage of the welding-type power supplied to the torch **16**. The control circuitry **30** may monitor the current and/or voltage of the arc **24** based at least in part on one or more sensors **36** within the wire feeder **14** and/or torch **16**. In some examples, a processor **35** of the control circuitry **30** may determine and/or control the arc length or electrode extension based at least in part on feedback from the sensors **36**. The processor **35** may determine and/or control the arc length or electrode extension utilizing data (e.g., algorithms, instructions, operating points) stored in a memory **37**. The data stored in the memory **37** may be received via the operator interface **28**, a network connection, or preloaded prior to assembly of the control circuitry **30**.

[0050] FIG. **2** is an example welding torch **16** that may be used in and/or with the example welding system of FIG. **1**. The torch **16** includes a handle **38** attached to a trigger **22**. The trigger **22** may be actuated to initiate a weld (and/or other welding-type operation). At a rear end **40**, the handle **38** is coupled to a cable **42** where welding consumables (e.g., the electrode **18**, the shielding gas, and so forth) are supplied to the weld. Welding consumables generally travel through the handle **38** and exit at a front end **44**, which is disposed on the handle **38** at an end opposite from the rear end **40**.

[0051] The torch **16** includes a neck **46** (e.g., a gooseneck) extending out of the front end **44** of the handle **38**. As such, the neck **46** is coupled between the handle **38** and a welding nozzle **48**. As should be noted, when the trigger **22** is pressed or actuated, welding wire (e.g., electrode **18**) travels through the cable **42**, the handle **38**, the neck **46**, and the welding nozzle **48**, so that the welding wire extends out of the front end **50** (i.e., torch tip) of the welding nozzle **48**. The handle **38** is secured to the neck **46** via fasteners **52** and **54**, and to the cable **42** via fasteners **52** and **54**. The welding nozzle **48** is illustrated with a portion of the welding nozzle **48** removed to show the electrode **18** extending out of a contact tip **56** that is disposed within the welding nozzle **48**. While the example torch **16** illustrated in FIG. **2** is designed for welding by a human operator, one or more torches designed for use by a robotic welding system may alternatively, or additionally, be used with the welding system of FIG. **1**. For example, the torch **16** may be modified to omit the trigger **22**, may be adapted for water cooling, etc.

[0052] FIGS. **3**A and **3**B show an example neck and nozzle assembly **300** that may be used with the welding torch **16** of FIG. **2**, and/or welding-type system **10** of FIG. **1**. In the example of FIGS. **3**A and **3**B, the neck and nozzle assembly **300** includes a nozzle assembly **348**, an insulator cap **304** (e.g., an electrical insulator cap), a contact tip **356**, and a neck assembly **346** (e.g., a MIG and/or GMAW welding neck assembly). FIG. **4**A shows an exploded view of the example neck and nozzle assembly **300** of FIGS. **3**A and **3**B. As shown in the example of FIG. **4**A, the neck and nozzle assembly **300** further includes a gas diffuser **500** having an O-ring **502**. FIG. **4**A also shows a liner assembly **349**, which is part of the neck assembly **346**. In the example of FIG. **4**A, the components of the neck and nozzle assembly **300** are centered about (and/or around, along, etc.) a longitudinal axis **301**.

[0053] FIGS. **4**B and **4**C show exploded views of the example neck assembly **346** and nozzle assembly **348**, respectively. In the example of FIG. **4**B, the neck assembly **346** includes an outer neck armor **350**, a neck insulation **352** (e.g., electrical neck insulation), a neck inner portion **354** (e.g., an electrically conductive neck inner portion), and a liner assembly **349**. In the example of FIG. **4**C, the outer neck armor **350**, neck insulation **352**, and neck inner portion **354** are generally cylindrical and hollow, with cylindrical bores centered about the axis **301** extending through the components.

[0054] The neck inner portion **354** may be comprised of an electrically conductive material. In the example of FIGS. **4**A and **4**B, the neck inner portion **354** includes screw threads **399** (e.g., a custom double start stub ACME screw thread) configured for coupling to the gas diffuser **500**, as will be discussed further below. The neck inner portion **354** further includes a base **355** to the rear of the screw threads **399**. A nose **362** of the neck inner portion **354** includes a locking taper **360**, forward of the screw threads **399**. The locking taper **360** may be configured to abut and/or engage with a complementary locking taper **560** of the gas diffuser **500** to assist with this coupling. The neck insulation **352** provides electrical (and/or thermal) insulation between the neck inner portion **354** and the outer neck armor **350**, and may be formed of an electrically insulating material. The liner assembly **349** provides a conduit through which the electrode **18** may travel from the torch handle **38** to the front end **50** of the nozzle assembly **348**. The outer neck armor **350**, neck insulation **352**, neck inner portion **354**, and liner assembly **349** include bores extending through their center which are aligned along a longitudinal axis **301**. When the neck assembly **346** is fully assembled, the liner assembly **349** is positioned within the bore of the neck inner portion **354**, the neck inner portion **354** is positioned within the bore of the neck insulation **352**, and the neck insulation **352** is positioned within the bore of the outer neck armor **350**.

[0055] The nozzle assembly **348** includes a nozzle body **306**, a nozzle insulator **308** (e.g., a nozzle electrical insulator), and a tip-retention device **310** (e.g., a nozzle insert). The tip-retention device **310** helps to retain the contact tip **356** within the nozzle assembly **348**. The nozzle insulator **308** provides electrical (and/or thermal) insulation within the nozzle assembly **348**, and may be formed of an electrically insulating material. In the example of FIG. **4**C, the nozzle body **306** and nozzle insulator **308** are generally cylindrical. In the example of FIG. **4**C, the nozzle body **306**, nozzle insulator **308**, and tip-retention device **310** include a bore centered about a common longitudinal axis **301**. When the nozzle assembly **348** is assembled, the tip-retention device **310** is positioned within the nozzle insulator **308**, and the nozzle insulator **308** is positioned within the nozzle body **306**.

[0056] In some examples, the tip-retention device **310** can be, for example, a nozzle insert or a nozzle addition that is crimped into or outside of the nozzle body **306**. In other examples, the tip-retention device **310** can be an integral part of the nozzle body **306** or the nozzle assembly **348**. In some examples, the tip-retention device **310** is configured to provide clearance for gas flow by providing channels through a side wall to direct gas inwardly towards the contact tip **356**. The inward gas flow is directed at the contact tip **356** which provides a cooling effect on the contact tip. Inwardly facing gas channels (e.g., radial channels) resist spatter collection in comparison to forward-facing gas holes (e.g., axial channels). Spatter can be removed from the front face of the tip-retention device **310** using a nozzle reamer or welpers. In some examples (e.g., as shown in FIG. **8**B), the tip-retention device **310** includes a taper **321** that is configured to lock the contact tip **356** in place. In some examples, the contact tip **356** has a locking taper **320** that is configured to engage with the taper **321** in the tip-retention device **310** to maintain concentricity and conductivity. In some examples, the contact tip **356** also includes a stepped profile or a backwards facing taper to engage with the gas diffuser **500** and to assist in maintaining concentricity and alignment with liner assembly **349**.

[0057] In some examples, the contact tip **356** may be threaded or threadless. If threadless, no tool may be necessary to insert the contact tip **356** into the neck and nozzle assembly **300**, for example. In some examples, the contact tip **356** can be secured with the use of a tool. In some examples, the contact tip may provide a consumable electrode **18**. In some examples, the contact tip **356** may be replaced with an electrode **18**, such as in examples where the welding torches do not provide and/or use a consumable electrode **18**.

[0058] In some examples, the neck and nozzle assembly **300** further includes a gas diffuser **500** (e.g., an electrically conductive gas diffuser). FIGS. **5**A-**5**C show various views of the gas diffuser

500. In the examples of FIGS. **5**A-**5**C, the gas diffuser **500** has a nose **504** with gas holes **506** configured to allow gas (e.g., shielding gas) to diffuse through into an interior of the nozzle assembly **348**. The nose **504** includes wrench flats **508** configured to allow a wrench to grip the nose **504** and/or tighten the gas diffuser **500** onto the neck assembly **346** (e.g., the neck inner portion **354**). In the examples of FIGS. **5**A-**5**C, the gas diffuser **500** further includes a generally cylindrical base **510**, and a generally cylindrical midsection **512** that connects the nose **504** to the base **510**. The outer diameter of the base **510** is larger than the outer diameter of the midsection **512** and the nose **504**. The midsection **512** has an outer diameter that is smaller than that of the base **510**, and larger than the outer diameter of the nose **504**. The nose **504** has an outer diameter that is smaller than the outer diameters of the base **510** and the midsection **512**. An O-ring **502** is fitted within an annular crevice on the outside of the base **510**. The O-ring **502** may be configured to create a gas seal between the gas diffuser **500** and the nozzle insulator **308**. The gas seal created by the O-ring may be configured to preclude gas from flowing backwards from the gas diffuser **500** towards the handle **38** of the welding torch **16**.

[0059] In some examples, the gas diffuser **500** includes a seat **514** that is configured to receive the contact tip **356**. A backwards facing taper of the contact tip **356** may interface and/or engage with a corresponding taper of the seat **514** of the gas diffuser **500**. While the rear end of the contact tip **356** is disposed against the seat **514** of the gas diffuser **500**, the tip-retention device **310** of the nozzle assembly **348** keeps the contact tip **356** in place. In some examples, a locking taper **320** (e.g., a forward-facing locking taper) of the contact tip **356** engages with a corresponding locking taper **321** of the tip-retention device **310** of the nozzle assembly **348**. In some examples, the locking taper **320** is disposed closer to a rear end of the contact tip **356** than a front end of the contact tip **356**. The contact tip **356** may be locked in place between the locking taper **321** of the tip-retention device **310** and the seat **514** of the gas diffuser **500**. In some examples, the contact tip **356** does not need its own threads to be locked in place.

[0060] In the examples of FIGS. **5**A-**5**C, the gas diffuser **500** further includes threaded grooves **599**. The threaded grooves **599** may be configured to complement and/or engage screw threads **399** of the neck assembly **346**. In the example of FIGS. **5**A-**5**C, the threaded grooves **599** are formed on an interior surface of the base **510**. The threaded grooves **599** travel (and/or extend) around the interior surface of the base **510** to form a spiral and/or corkscrew pattern. In some examples, the threaded grooves **599** may be double start threaded grooves **599**. In the example of FIGS. **5**A-**5**D, the threaded grooves **599** are double start stub ACME threaded grooves. As the threaded grooves **599** are double start, the threaded grooves **599** comprise two separate grooves: a first groove **599**A and a second groove **599**B (separated by protruding threads **599**C). The threaded grooves **599** have a grooved engagement distance **592**, a grooved lead distance **594**, and a grooved pitch distance **596**. The grooved engagement distance **592** is the total axial length of the threaded grooves **599**, comprising the total axial distance complementary screw thread **399** can travel when engaging the threaded grooves **599**. The grooved lead distance **594** comprises an axial distance between two of the same adjacent grooves (e.g., the distance between adjacent first grooves **599**A), and the axial distance traversed during one complete revolution (and/or turn) within a groove. The grooved pitch distance **596** comprises an axial distance between any two adjacent grooves (e.g., adjacent first grooves **599**A and/or second grooves **599**B). The threaded grooves **599** further comprise a groove major diameter **598**C (e.g., the diameter of the gas diffuser **500** at the threads **599**C), a groove minor diameter **598**A (e.g., the diameter of the gas diffuser **500** at the grooves **599**A, **599**B), and a groove pitch diameter **598**B (average/mean of groove major diameter **598**C and groove minor diameter **598**A).

[0061] The neck assembly **346** may include screw threads **399** that are complementary to, and/or configured to engage, the threaded grooves **599** of the gas diffuser **500**. More particularly, the neck inner portion **354** may include screw threads **399** complementary to the threaded grooves **599** of the gas diffuser **500**. The screw threads **399** travel (and/or extend) around an outer surface of the neck assembly **346** to form a spiral and/or corkscrew pattern. In the examples of FIG. **6**A-**6**D, the screw threads **399** are double start screw stub ACME screw threads, comprising a first screw thread **399**A (configured to engage the first threaded groove **599**A) and a second screw thread **399**B (configured to engage the second threaded groove **599**B). The first screw thread **399**A is separated from the second screw thread **399**B by recessed grooves **399**C. The screw threads **399** have a threaded engagement distance **392**, a threaded lead distance **394**, and a threaded pitch distance **396**. The threaded engagement distance **392** is the total axial length of the screw threads **399**, comprising the total axial distance the screw threads **399** can travel when engaging the threaded grooves **599**. The threaded lead distance **394** comprises an axial distance between the same adjacent threads (e.g., the distance between adjacent first screw threads **399**), and the axial distance traversed during one complete revolution (and/or turn) of a screw thread within a complementary groove. The threaded pitch distance **396** comprises an axial distance between any two adjacent threads (e.g., between adjacent first screw threads **399**A and/or second screw threads **399**B).

[0062] The screw threads **399** further comprise a thread major diameter **398**F (e.g., diameter of the neck inner portion **354** at the screw threads **399**A, **399**B), a thread minor diameter **398**D (e.g., diameter of the neck inner portion **354** at the recessed grooves **399**C), and a thread pitch diameter **398**E (e.g., an average/mean diameter). In some examples, the groove major diameter **598**C, groove minor diameter **598**A, and/or groove pitch diameter **598**B may be slightly greater than the thread major diameter **398**F, thread minor diameter **398**D, and/or thread pitch diameter **398**E, respectively, in order to accommodate the screw threads **399** within the threaded grooves **599**. In some examples, the groove major diameter **598**C, groove minor diameter **598**A, and/or groove pitch diameter **598**B may be approximately equal to the thread major diameter **398**F, thread minor diameter **398**D, and/or thread pitch diameter **398**E, respectively. The base **355** of the inner neck portion **354** may further comprise a base diameter **358**.

[0063] FIGS. **8**A and **8**B show a cross-section view and a partial cutaway perspective view, respectively, of the example neck and nozzle assembly **300**. In the example of FIGS. **8**A and **8**B, the gas diffuser **500** is coupled to the nozzle assembly **348** and neck assembly **346**. The gas diffuser **500** may be attached to the nozzle assembly **348** via external threads (not shown) of the gas diffuser **500** mating with internal threads (not shown) of the nozzle assembly **348**, and/or through a frictional fit. In the example of FIGS. **8**A and **8**B, the gas diffuser **500** is also attached to the tip-retention device **310** (e.g., a nozzle insert) of the nozzle assembly **348**, such as through a frictional fit. The liner assembly **349** further extends into the gas diffuser **500**, terminating adjacent the seat **514** so as to provide a route for the electrode **18** to move through the neck assembly **346** and gas diffuser **500** into the contact tip **356**. The contact tip **356** is retained within the neck and nozzle assembly **300** by a frictional fit with the gas diffuser **500** (e.g., through the seat **514**) and the tip-retention device **310**. In the example of FIGS. **8**A and **8**B, the gas diffuser **500** is attached to the welding neck assembly **346** via threaded grooves **599** of the gas diffuser **500** mating with screw threads **399** of the welding neck assembly **346**.

[0064] In operation, the gas diffuser **500** may be screwed onto the welding neck assembly **346** using the screw threads **399** of the welding neck assembly **346** and/or threaded grooves **599** of the gas diffuser **500**. For example, the screw threads **399** of the welding neck assembly **346** may mate, interface, and/or engage with the threaded grooves **599** of the gas diffuser **500** to secure the gas diffuser **500** to the neck assembly **346**. In some examples, the threaded grooves **599** of the gas diffuser **500** may be brought into contact with the screw threads **399** of the neck assembly **346**, and the gas diffuser **500** may be turned (and/or rotated, twisted, moved, etc.) to connect the gas diffuser **500** to the neck assembly **346**. In some examples, the turning of the gas diffuser **500** with respect to the welding neck assembly **346** may be achieved manually without any tools (e.g., a tool-less option) or may be turned using a tool (e.g., wrench, etc.) that is applied to, for example, the wrench flats **508**. The wrench flats **508** may provide a gripping surface through which a tool (and/or hand) may tighten the gas diffuser **500** onto the welding neck assembly **346**.

[0065] In some examples, the screw threads **399** and/or threaded grooves **599** may be configured such that the gas diffuser **500** may be coupled to and/or removed from the with approximately 1.5 turns or less. For example, the total threaded engagement distance **392** of the screw threads **399** may be approximately 1.5 times the threaded lead distance **394**, or less (e.g. 1.3, 1.4, 1.5, 1.6, or 1.7 turns). In some examples, the total grooved engagement distance **592** of the threaded grooves **599** may be approximately 1.5 times the grooved lead distance **594** (or less). As the threaded lead distance **394** (and/or grooved lead distance **594**) is the axial distance traveled by the screw threads **399** (and/or threaded grooves **599**) over one turn of the neck assembly **346** (and/or gas diffuser **500**), the neck assembly **346** (and/or gas diffuser **500**) may travel the full threaded engagement distance **392** (and/or total grooved engagement distance **592**) in approximately 1.5 turns (or less) when the total threaded engagement distance **392** (and/or total grooved engagement distance **592**) is 1.5 times (or less) the threaded lead distance **394** (and/or grooved lead distance **594**).

[0066] In some examples, the screw threads **399** and/or threaded grooves **599** may be configured such that the gas diffuser **500** may be coupled to and/or removed from the with approximately 2 turns or less (e.g., 1.8, 1.9, 2, 2.1, or 2.2 turns). For example, the total threaded engagement distance **392** of the screw threads **399** may be approximately 2 times the threaded lead distance **394** (or less). In some examples, the total grooved engagement distance **592** of the threaded grooves **599** may be approximately 2 times the grooved lead distance **594** (or less). As the threaded lead distance **394** (and/or grooved lead distance **594**) is the axial distance traveled by the screw threads **399** (and/or threaded grooves **599**) over one turn of the neck assembly **346** (and/or gas diffuser **500**), the neck assembly **346** (and/or gas diffuser **500**) may travel the full threaded engagement distance **392** (and/or total grooved engagement distance **592**) in approximately 2 turns (or less) when the total threaded engagement distance **392** (and/or total grooved engagement distance **592**) is 2 times (or less) the threaded lead distance **394** (and/or grooved lead distance **594**).

[0067] In some examples, the total grooved engagement distance **592** may be approximately equal to the total threaded engagement distance **392**. In some examples, the total grooved engagement distance **592** may be different from the total threaded engagement distance **392**. In some examples, the threaded lead distance **394** may be approximately equal to the grooved lead distance **594**, so that the screw threads **399** properly engage with the threaded grooves **599**. In some examples, the threaded pitch distances **396** may be approximately half the threaded lead distances **394**, and/or the grooved pitch distances **596** may be approximately half the grooved lead distances **594**.

[0068] The table **700** of FIG. **7** shows various potential example dimensions (in inches) of the screw threads **399** on the welding neck assembly **346** and mating threaded grooves **599** in the gas diffuser **500** according to the present disclosure. More particularly, the table **700** shows potential major, minor, and pitch diameters for the screw threads **399** and/or threaded grooves **599**. The table **700** also show potential threaded lead distances **394**, threaded pitch distances **396**, grooved lead distances **594**, and/or grooved pitch distances **596**. The second column (0.425-16) of the table **700** shows potential dimensions for a neck assembly **346** having a base diameter **358** of 0.438 inches, and a threaded engagement distance **392** (and/or grooved engagement distance **592**) of 0.185 inches. The third column (0.527-14) of the table **700** shows potential dimensions for a neck assembly **346** having a base diameter **358** of 0.540 inches, and a threaded engagement distance **392** (and/or grooved engagement distance **592**) of 0.215 inches. The “0.425” in “0.425-16” (and/or “0.527” in “0.527-14”) refers to the max thread major diameter **398**F. The “16” (and/or “14”) is the thread pitch, comprising the number of threads per inch. The dimensions in the table **700** may be used to customize the screw threads **399** and/or threaded grooves **599** contemplated by the present disclosure. In some examples, the dimensions and/or tolerances of the screw thread **399** and/or threaded groove **599** features (e.g., major, minor, and/or pitch diameters, lead and/or pitch distances, locking tapers, etc.) may slightly alter the number of turns required.

[0069] FIG. **9**A shows a cross-sectional view of an example gas diffuser **500** and/or welding neck assembly **346** in which the screw threads **399** and/or threaded grooves **599** are used with a face-to-face bearing surface version of the gas diffuser **500** and welding neck assembly **346**. In the example of FIG. **9**A, the gas diffuser **500** includes a bearing surface **561** rather than a locking taper **560**, and the neck inner portion **354** of the neck assembly **346** includes a complementary bearing surface **361** rather than the locking taper **360**. The bearing surface **561** of the gas diffuser **500** abuts (and/or interacts/interfaces with, engages, etc.) the complementary bearing surface **261** of the neck inner portion **354** to assist in locking the gas diffuser **500** and the welding neck assembly **346** with respect to each other. FIG. **9**B shows a cross-sectional view of an example gas diffuser **500** and/or welding neck assembly **346** in which the screw threads **399** and/or threaded grooves **599** are custom double start ANSI-style threads/grooves, rather than ACME style threads/grooves.

[0070] While the above disclosure refers to screw threads **399** of the neck assembly **346** and threaded grooves **599** of the gas diffuser **500**, in some examples the neck assembly **346** may include the threaded grooves **599** and the gas diffuser **500** may include the screw threads **399**. It is further understood that the screw threads **399** include grooves between threads, and the threaded grooves **599** include threads between grooves, and that all of these features may interact (and/or engage, interface, interlock, abut, etc.) when coupling the gas diffuser **500** to the neck assembly **346**.

[0071] FIGS. **10**A and **10**B show an example gas diffuser **500** and neck inner portion **354** having a different quick connect configuration. FIG. **10**A shows a side view of an example neck inner portion **354** of the example welding neck assembly **346** in which the screw threads **399** are replaced with keyed protrusions **389**. FIG. **10**B shows a rear end view of an example gas diffuser **500** in which the threaded grooves **599** are replaced with keyways **588** configured to receive the protrusions **389**, and channels **589** with which the protrusions **389** may interface, and/or in which the protrusions **389** may move. In some examples, the gas diffuser **500** may instead be formed with the protrusions **389**, and the neck inner portion **354** may be formed with the keyways **588** and/or channels **589**.

[0072] In the example of FIG. **10**A, the keyed protrusions **389** are formed on the neck inner portion **354** of the welding neck assembly **346**, between the nose **362** and the base **355**. The protrusions **389** are substantially aligned along the neck inner portion **354**. In some examples, the protrusions **389** may instead be offset. In the example of FIG. **10**A, there are two protrusions **389**, though in some examples there may be more or less than two protrusions **389** (e.g., one protrusion, three protrusions, four protrusions, etc.).

[0073] In the example of FIG. **10**B, the keyways **588** and/or channels **589** are formed on/in the interior surface of the base **355** of the gas diffuser **500**. The keyways **588** are formed on/in the interior surface of the base **355**, beginning at the rear of the gas diffuser **500** and extending towards the nose **362**, so as to provide an avenue for the keyed protrusions **389** to reach the channels **589**. In some examples, the keyways **588** may be configured to be a length sufficient to allow the locking tapers **360**, **560** to abut (and/or interface, engage, etc.) by the time the protrusions **389** reach the channels **589**. In some examples (such as where the channels **589** are formed helically, for example), the keyways **588** may be configured to be a length that requires the protrusions **389** to be turned within the channels before the locking tapers **360**, **560** abut (and/or interface, engage, etc.).

[0074] While two keyways **588** are shown in the example of FIG. **10**B, in other examples there may be more or less keyways. The keyways **588** may be configured to complement the configuration of the keyed protrusions **389** (e.g. with respect to spacing, size, etc.). In some examples, there may be more keyways **588** than keyed protrusions **389**. If not aligned properly with the keyways **588**, the protrusions **389** may abut the gas diffuser **500** and prevent the protrusions **389** from reaching the channels **589**, and/or prevent the gas diffuser **500** and welding neck assembly **346** from coupling.

[0075] In the example of FIG. **10**B, the channels **589** begin at the axial ends of the keyways **588**. The channels **589** comprise arcs centered about the longitudinal axis **301** of the gas diffuser **500**, and extending around the interior surface of the gas diffuser **500** in a radial (and/or circumferential, circular, radially tangential, etc.) direction. In the example of FIG. **10**B, there are two channels **589** that extend approximately a quarter of the way around the circumference of the gas diffuser **500**. In some examples, there may be more or fewer channels **589**. In some examples, the channels **589** may extend an eight of the way around the circumference, halfway around the circumference, all the way around the circumference (such as if there is only one keyway, or multiple offset keyways, for example), and/or more than all the way around the circumference. In some examples, the channels **589** may be formed helically. In some examples, the channels **589** may be formed more circularly. The channels **589**, like the keyways **588**, may be configured to complement the configuration of the keyed protrusions **389**.

[0076] In operation, the protrusions **389** may be aligned with the keyways **588** when connecting the gas diffuser **500** to the welding neck assembly **346**, and then moved within the channels **589** (e.g., through a twisting and/or turning motion of the gas diffuser **500** and/or neck inner portion **354**) to lock the gas diffuser **500** to the welding neck assembly **346**. In the example of FIGS. **10**A and **10**B, the connection may be made in a quarter turn or less. Once turned, the keyed protrusions **389** may be out of alignment with the keyways **588**, thus preventing (and/or resisting) decoupling. In some examples, an axial width of the channels **589** may shrink (and/or narrow, reduce, change, etc.) as the channels **589** extend radially around the gas diffuser **500**. This change in width may squeeze the protrusions **389** between the sidewalls of the channels **589** when the gas diffuser **500** and/or neck inner portion **354** is rotated, thereby forming a frictional engagement that resists decoupling.

[0077] In some examples, a spring may be included within gas diffuser **500**. The spring may be configured to provide an axial spring force that pushes against the protrusions **389** when the protrusions **389** enter the keyway **588** and/or the channels **589**. Thus, the protrusions **389** may compress the spring when pushed into the keyways **588** to reach the channels **589**, and the spring force may push the protrusions **389** back against the walls of the channels **589** once the protrusions are rotated out of alignment with the keyways **588**.

[0078] While the present apparatuses, systems, and/or methods have been described with reference to certain implementations, it will be understood by those skilled in the art that various changes may be made and equivalents may be substituted without departing from the scope of the present apparatuses, systems, and/or methods. In addition, many modifications may be made to adapt a particular situation or material to the teachings of the present disclosure without departing from its scope. Therefore, it is intended that the present apparatuses, systems, and/or methods not be limited to the particular implementations disclosed, but that the present apparatuses, systems, and/or methods will include all implementations falling within the scope of the appended claims.

Claims

1. A gas diffuser for a welding torch, comprising: a rear diffuser portion; a front diffuser portion configured for connection with a contact tip of the welding torch; a middle diffuser portion connecting the rear diffuser portion and front diffuser portion; a passage that extends from a rear end of the rear diffuser portion, through the rear diffuser portion, middle diffuser portion, and front diffuser portion, to a front end of the front diffuser portion, wherein an inner wall of the rear diffuser portion encircles a portion of the passage; and a threaded groove formed in or on the inner wall of the rear diffuser portion, the threaded groove comprising a first groove and a second groove, the first groove not being continuous with the second groove, and an axial distance between the first groove and the second groove being approximately 0.0714 inches or 0.0625 inches, the threaded groove being configured to interface with a protrusion of a neck assembly of the welding torch to connect the gas diffuser to the neck assembly in approximately 1.5 turns or less.

2. The gas diffuser of claim 1, wherein the passage has a diameter of between 0.5499 inches and 0.4846 inches, or between 0.4464 inches and 0.3875 inches, at the threaded groove.

3. The gas diffuser of claim 1, wherein a groove diameter of the passage at the first groove is between 0.5499 inches and 0.5375 inches, or between 0.4464 inches and 0.4350 inches.
4. The gas diffuser of claim 1, wherein the first groove is adjacent the second groove.
5. The gas diffuser of claim 4, wherein the first groove and second groove are separated by a protruding thread.
6. The gas diffuser of claim 5, wherein a thread diameter of the passage at the protruding thread is between 0.4882 inches and 0.4846 inches, or between 0.3906 inches and 0.3875 inches.
7. The gas diffuser of claim 1, wherein the threaded groove is a double start groove.
8. The gas diffuser of claim 1, wherein the threaded groove is based on a custom double-start stub ACME thread.
9. The gas diffuser of claim 1, wherein the gas diffuser further comprises a middle inner wall of the middle diffuser portion, the middle inner wall having a taper such that a front diameter of the passage at the front diffuser portion is less than a rear diameter of the passage at the rear diffuser portion, the taper of the middle inner wall being configured to engage with a complementary taper of the neck assembly of the welding torch.
10. The gas diffuser of claim 1, wherein the rear diffuser portion is configured for connection with the neck assembly of the welding torch.
11. A gas diffuser for a welding torch, comprising: a rear diffuser portion; a front diffuser portion configured for connection with a contact tip of the welding torch; a middle diffuser portion connecting the rear diffuser portion and front diffuser portion; a passage that extends from a rear end of the rear diffuser portion, through the rear diffuser portion, middle diffuser portion, and front diffuser portion, to a front end of the front diffuser portion, wherein an inner wall of the rear diffuser portion encircles a portion of the passage; and a threaded groove formed in or on the inner wall of the rear diffuser portion, the threaded groove comprising a first groove and a second groove, the first groove not being continuous with the second groove, and an axial distance between the first groove and the second groove being approximately 0.0714 inches or 0.0625 inches, the threaded groove configured for connection with a complementary screw thread of a neck assembly of the welding torch, the threaded groove having a total axial length that is approximately 1.5 times as long as a lead length of the first groove.
12. The gas diffuser of claim 11, wherein the passage has a diameter of between 0.5499 inches and 0.4846 inches, or between 0.4464 inches and 0.3875 inches, at the threaded groove.
13. The gas diffuser of claim 11, wherein a groove diameter of the passage at the first groove is between 0.5499 inches and 0.5375 inches, or between 0.4464 inches and 0.4350 inches.
14. The gas diffuser of claim 11, wherein the first groove is adjacent the second groove.
15. The gas diffuser of claim 14, wherein the first groove and second groove are separated by a protruding thread.
16. The gas diffuser of claim 15, wherein a thread diameter of the passage at the protruding thread is between 0.4882 inches and 0.4846 inches, or between 0.3906 inches and 0.3875 inches.
17. The gas diffuser of claim 11, wherein the threaded groove is a double start groove, or the threaded groove is based on a custom double-start stub ACME thread.
18. The gas diffuser of claim 11, wherein the gas diffuser further comprises a middle inner wall of the middle diffuser portion, the middle inner wall having a taper such that a front diameter of the passage at the front diffuser portion is less than a rear diameter of the passage at the rear diffuser portion, the taper of the middle inner wall being configured to engage with a complementary taper of the neck assembly of the welding torch.
19. The gas diffuser of claim 11, wherein the rear diffuser portion is configured for connection with the neck assembly of the welding torch.
20. The gas diffuser of claim 11, wherein the passage is centered around an axis of the gas diffuser, the axis defining an axial direction, the total axial length comprising a total distance in the axial direction that the complementary screw thread can travel when engaging the threaded groove, and

the lead length of the first groove comprising a distance in the axial direction between adjacent portions of the first groove.